

Enhancing Radiography Education through Co-located Mixed Reality with Tracker-based X-ray Imaging Simulation

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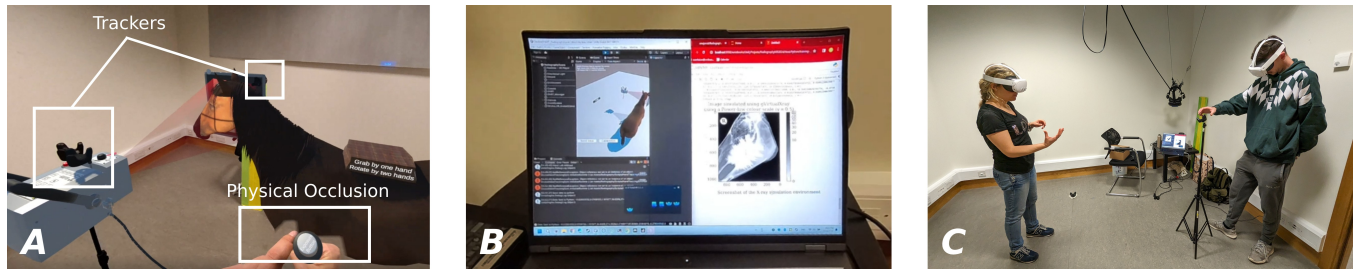


Figure 1: (A) User's view when taking X-ray; (B) Real-time X-ray simulation on the host PC; (C) Expert review setup.

ABSTRACT

This poster proposes a co-located collaborative Mixed Reality (MR) teaching system to enhance Veterinary Radiography training. By integrating real X-ray equipment with virtual representations of horses, the system simulates radiographic procedures effectively. The cost-effective setup combines real equipment with virtual annotations, such as collimation beams, to achieve synchronized virtual objects and generate real-time radiography. This innovative approach enables instructors to demonstrate equine techniques and assess students' proficiency more effectively. An expert review study was conducted to gather feedback and evaluate user experience related to this novel teaching approach.

CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality**; **Heuristic evaluations**; **Collaborative interaction**.

KEYWORDS

Mixed reality, Co-location, Collaborative, Radiography, Training.

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1 INTRODUCTION

While learning how to perform radiographic techniques, veterinary students often struggle with the positioning of the radiographic tube or the image detector, leading to unnecessary repetitions of image acquisition and potential exposure to harmful radiation for both patients and staff. To address this, strict radiation safety protocols are in place, but these limit the number of students allowed in the room during equine radiography, hindering training effectiveness. Furthermore, sedating animals purely for teaching purposes raises ethical concerns, limiting the time dedicated to radiographic technique training. Fortunately, virtual simulation offers solutions to these challenges, improving both performance and training efficiency. Immersive virtual environments, including Head-mounted-display (HMD) based Extended Reality (XR), have been shown to enhance radiography training [Bannister et al. 2019; O'Connor et al. 2021; Xu et al. 2023]. Real-time radiograph simulation systems combined with XR technologies have been found to improve trainee performance and reduce procedural time [Killeen et al. 2023; Sujar et al. 2022]. Tracker-based collaborative MR systems replicate clinical procedures, allowing students to practice safely without radiation exposure. An expert review study evaluated the proposed system, gathering feedback on simulation quality and user experience.

2 SYSTEM SETUP

Prior research presented a co-located collaborative MR environment employing multiple mobile VR headsets to facilitate equine radiography training processes [Xu et al. 2023]. Possibilities for system enhancement were identified: 1) The only requirement for virtual visualization pertains to the animals. 2) Integration of authentic equipment into the training protocol would closely simulate real-world scenarios. 3) Real-time X-ray radiography generation.

The system comprises two main components: the host and users, interacting within both physical and virtual environments. The

host component consists of a tracking system and a real-time X-ray simulation system, while the user component encompasses synchronization procedures, physical interactions, and interactions within XR. Calibration within both components relies on an anchor in the physical world. The host component tracks the transforms of physical trackers wirelessly synchronized with the physical world (Figure 1-A), transmitting this data through the network to serve as anchors for virtual annotations within the virtual environment. Users, interacting with both physical and virtual elements, can manipulate real imaging equipment while engaging with virtual annotations. The Depth API was enabled to make virtual objects occluded by real objects and appear to be embedded in the environment (Figure 1-A). Upon activation of the exposure button, all necessary data is transmitted to the host's real-time X-ray simulation system. This system then generates simulated radiographs based on the provided information (Figure 1-B).

3 EXPERT REVIEW

The study enrolled eight veterinary experts (Figure 1-C). Prior to each session, a technician researcher (XX) prepared the environment, ensuring synchronization and calibration, while another researcher (AP) provided a system introduction and demonstrated basic interactions. After familiarizing themselves with the system, participants performed equine techniques as instructed. Each participant was asked to produce at least two X-ray images of the equine head, which were assessed visually for quality. Subsequently, participants completed a questionnaire created using user experience questionnaire plus (UEQ+), Mixed Reality experience questionnaire (MREQ), and Virtual Reality sickness questionnaire (VRSQ), offering feedback and suggestions based on their session experience.

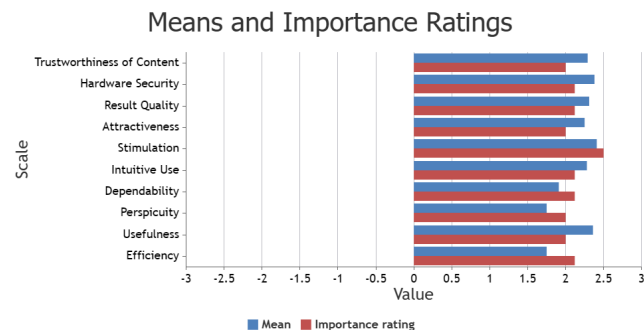


Figure 2: Mean and importance rating per scale.

The user experience evaluation utilized the UEQ+ modular framework, yielding predominantly high ratings across general aspects, averaging above 2 (Figure 2). The system was praised as impressive and interesting, with its usefulness highlighted for offering a safe and repeatable learning experience, especially beneficial for novice trainees requiring extensive hands-on practice. However, aspects like perspicuity and efficiency received lower rankings compared to other metrics. Scales related to the quality and accuracy of the X-ray imaging system received notably high ratings. Simulated X-ray images closely resembled real radiography, aligning with corresponding equipment and patient poses, while the

system workflow effectively mirrored real-world practices. This alignment suggests that the system is capable of providing novice trainees with a comprehensive and systematic simulation equivalent to real-world scenarios. Users' co-presence and presence within the co-located MR space were assessed using eight items from the MREQ questionnaire. The proposed system scored high (5.8) compared to a high point, indicating users found the virtual objects and other users convincing. Assessment with the VRSQ revealed no significant issues with the MR system. Participants reported minimal discomfort with no severe motion sickness symptoms.

Regarding image quality, participants emphasized the importance of simulating reality accurately in terms of exposure factors and positioning effects. Suggestions included adjusting X-ray machine power to correspond with simulated images and generating faulty images for inaccurate poses. Attention to detail, such as implementing a two-level exposure button, was also recommended. For patient presentations and clinical scenarios, participants noted the need for a variety of scenarios beyond the common supine position, including bent legs and pathologies requiring supplementary views, to enhance learning outcomes. Lastly, participants suggested adding virtual lead coats, blocks for foot X-rays, and a headrest to further enhance the system. Incorporating an actual X-ray machine into the teaching setting was also proposed to increase realism.

4 FUTURE WORK

Our intention is to comprehensively assess the system through a series of experiments targeting various aspects. We will recruit veterinary students engaged in clinical rotations to evaluate the system's efficacy in aiding their acquisition of radiography techniques and safety protocols. Furthermore, we will conduct a controlled comparative analysis with a fully virtual environment to determine the added value of incorporating real equipment and environments for teaching equine techniques. Additionally, we aim to objectively evaluate the system's practical assessment capabilities by comparing the performance of novice trainees and experienced specialists.

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